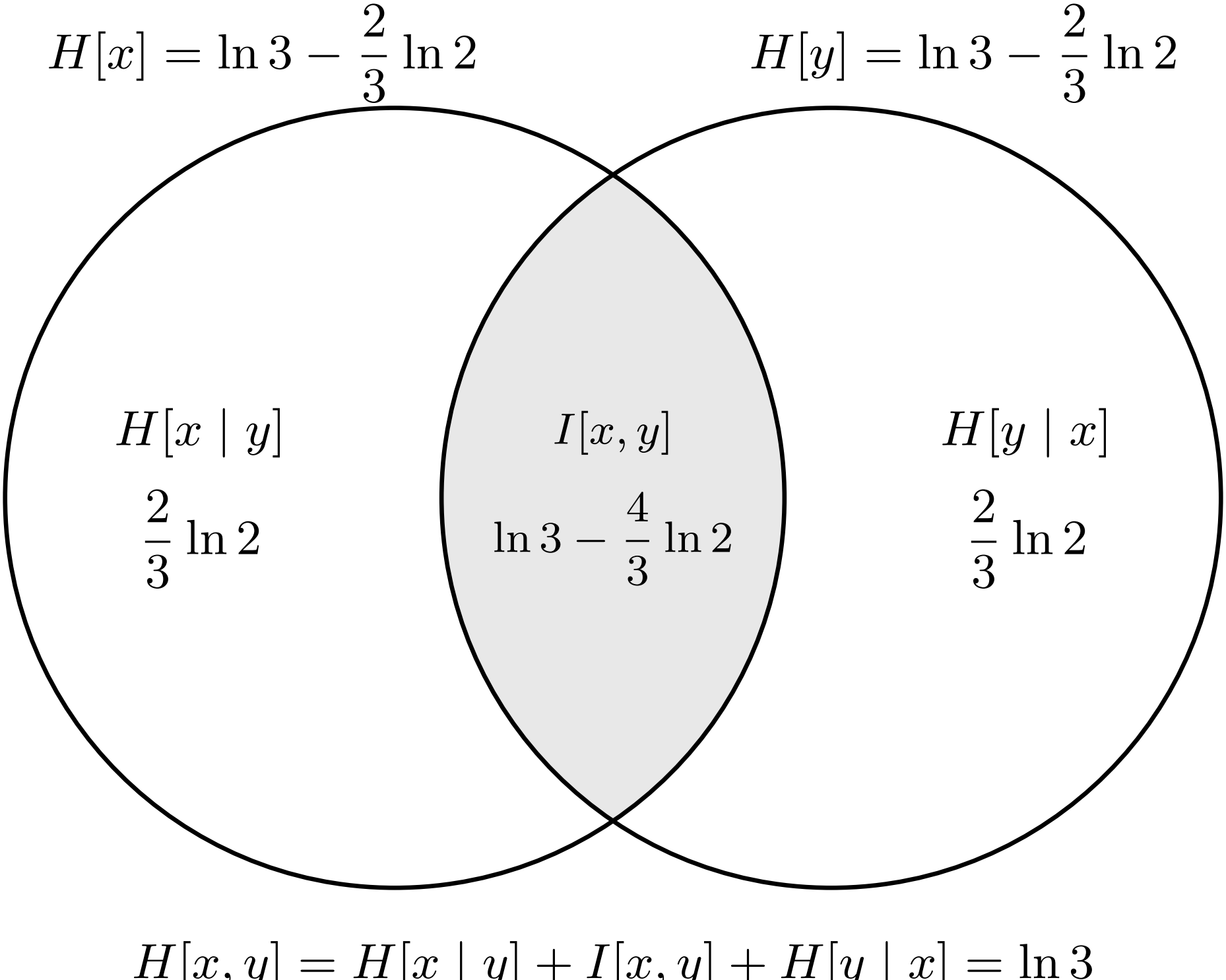

$$H[x, y] = H[x \mid y] + I[x, y] + H[y \mid x] = \ln 3$$